Root Cause Analysis Report

Part	MS-4505-001 Sheet Metal Bracket
Material / process	SPCC Steel / bending
Case	rca_903adb7f09f54e4182fc24002ced20cf
Date	2026-11-02 15:23:47
Status	AI proposal, pending expert review

1. Problem

Defect	Severity	Confidence	Detail
dent	major	84%	
dent	major	82%	
dent	major	68%	
dent	major	65%	
dent	major	58%	
dent	major	57%	
dent	major	36%	
dent	major	22%	
crack	critical	17%	
dent	major	16%	
dent	major	15%	
dent	major	13%	

2. Root cause

Cause: Material Hardness Variation

Reason: The detected critical crack and surface defects align with the documented effects of hardness variation in SPCC steel, which causes unpredictable behavior during bending operations [02cf0928-6739-4f7e-ae1b-f6740283f00f].

Effect: Cracking at bend radius, surface defects

Confidence: 76%

Supporting evidence
- **02cf0928-6739-4f7e-ae1b-f6740283f00f**: Material Inconsistency / Hardness Variation Root Cause Analysis Defect Types: crack, deformation, dimensional variation Material: SPCC steel, cold-rolled sheet metal Root Cause: Hardness variation in incoming material lot causes unpredictable springback behavior and susceptibility to cracking under forming operations. Effect: Cracking at bend radius, dimensional inaccuracy due to variable springba

3. Solution

Corrective actions (fix this occurrence)

1. Verify material certificates against specification for yield strength, tensile strength, and hardness
2. Conduct incoming material hardness testing

Preventive actions (stop it recurring)

1. Implement stricter incoming material quality control protocols
2. Audit material supplier quality management systems

Precautions

- Verify material certificates against specification for yield strength, tensile strength, and hardness
- Conduct incoming material hardness testing
- Implement stricter incoming material quality control protocols
- Audit material supplier quality management systems

4. Other causes considered

Rank	Cause	Confidence	Reason
2	Inadequate Process Control for Material Inconsistency	73%	The evidence suggests that material inconsistencies in SPCC steel lead to susceptibility to cracking during forming; if process parameters are not adjusted for specific lot properties, defects occur [02cf0928-6739-4f7e-ae1b-f6740283f00f].

5. Traceability

AI model: gemini:gemini-3.1-flash-lite. Expert edited the cause: no.
This analysis was drafted with AI assistance from the retrieved knowledge-base evidence. It must be reviewed and confirmed by a qualified quality expert before corrective action is taken.